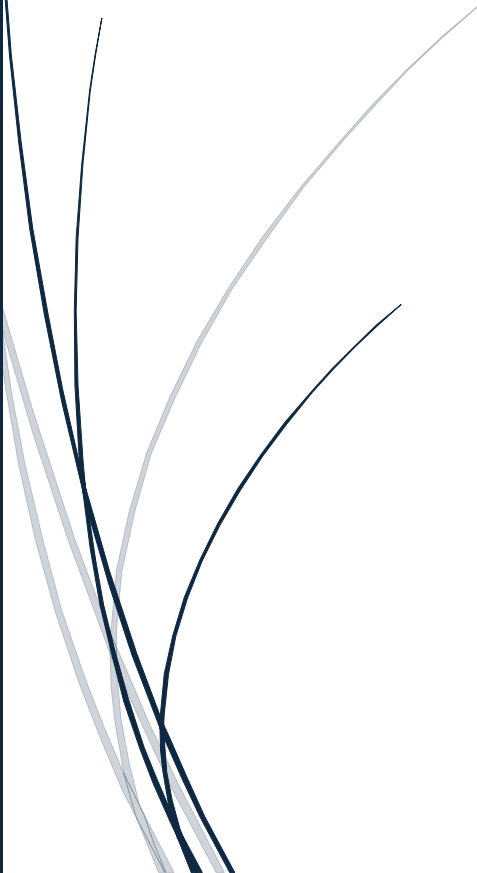




7/31/2025

Analyses & Report on a Cloud-Based Case Study

Module Code: UFCFKJ-15-M



Sourav Kumar Avi

1. Executive Summary	2
2. SWOT Analysis of Current IT Infrastructure and Services	2
2.1 Strengths	2
2.2 Weaknesses	2
2.3 Opportunities.....	2
2.4 Threats.....	3
3. SWOT Analysis of Proposed IT Infrastructure and Services	3
3.1 Strengths	3
3.2 Weaknesses	3
3.3 Opportunities.....	3
3.4 Threats.....	4
4. SaaS, PaaS & IaaS Cloud Computing Solutions	4
4.1 Software as a Service (SaaS)	4
4.2 Platform as a Service (PaaS).....	5
4.3 Infrastructure as a Service (IaaS).....	5
5. Recommended Commercial Provider and Cost Estimates	7
5.1 Recommended Provider: Microsoft Azure	7
5.2 Cost Estimates.....	7
6. INaaS & BPaaS Development Recommendations	8
6.1 Information as a Service (INaaS).....	9
6.2 Business Process as a Service (BPaaS).....	9
7. Sustainability Discussion	10
8. Conclusion	10
9. References	11

1. Executive Summary

Christian Volunteer Services (CVS), a longstanding UK charity, faces outdated IT systems, fragmented data, and high operational costs. This report proposes a cloud strategy using SaaS, PaaS, IaaS, INaaS, and BPaaS to improve efficiency, service delivery, governance, and sustainability. Microsoft Azure is the preferred provider, offering nonprofit discounts, integration, and sustainability benefits. The three-year strategy fits CVS's £10 million budget and supports its mission-driven growth.

2. SWOT Analysis of Current IT Infrastructure and Services

A structured SWOT analysis evaluates CVS's current IT setup, highlighting strengths, inefficiencies, and areas for improvement.

2.1 Strengths

Dedicated IT Workforce: 40 IT and 18 support/development staff offer strong internal capabilities (Assessment Brief, 2025).

Established Systems: Long-standing legacy databases store critical donor and volunteer records (6.4GB).

National Reach: A broad footprint (260 shops, 4,000 staff) supports scalable digital rollout.

Positive Work Culture: Favourable staff conditions foster digital adoption.

2.2 Weaknesses

High IT Costs: Annual spending is £25.7M, with £8.5M on licensing and £6.4M on hardware/networking.

Downtime: 15 lost working days/year equates to 1,500 staff days, lowering productivity.

Disjointed Tools: Five payroll systems and various spreadsheets increase inefficiencies.

Limited Digital Strategy: Absence of e-commerce and data-driven fundraising limits growth.

2.3 Opportunities

Cost Reduction via Cloud: Shifting to SaaS/IaaS can reduce CapEx and improve ROI (IDC, 2023).

CRM & Analytics: A centralised CRM boosts engagement and fundraising.

E-Commerce Potential: Cloud-enabled online sales can monetise high-value donations.

Data Insights: Cloud analytics enable evidence-based planning and reporting.

2.4 Threats

Change Resistance: Unionised staff may oppose digital transformation.

Cybersecurity Risks: Data migration poses privacy challenges (Ali et al., 2022).

Migration Disruptions: Unplanned downtime may affect daily operations.

Compliance Risk: Sensitive data requires strict GDPR adherence.

3. SWOT Analysis of Proposed IT Infrastructure and Services

This SWOT explores the benefits and trade-offs of CVS's proposed cloud-based IT environment.

3.1 Strengths

Scalability: Cloud services adapt to workload changes and reduce resource waste (Armbrust et al., 2010).

Cost Efficiency: OpEx-based models lower long-term costs (Microsoft, 2022).

System Centralisation: Cloud CRM, HRM, and payroll unify fragmented platforms.

Uptime Guarantees: Azure offers >99.9% SLAs, reducing outages.

Advanced Analytics: Tools like Power BI deliver real-time insights.

3.2 Weaknesses

Upfront Costs: Migration and training require initial investment.

Vendor Lock-In: Proprietary services may limit future flexibility (Wang et al., 2014).

Training Needs: Staff require upskilling for cloud adoption.

Internet Dependency: Shops and remote workers depend on stable connectivity.

3.3 Opportunities

BPaaS Innovation: Automates key processes like onboarding and donor follow-up.

Compliance Tools: Built-in features support GDPR and data governance.

Sustainable Operations: Migration to green data centres aligns with CVS's values (Google Cloud, 2023).

Smarter Fundraising: CRM-led campaigns improve donor retention.

3.4 Threats

Security Misconfigurations: Poor setup can expose sensitive data (Ali et al., 2022).

Cultural Resistance: Rapid change may affect morale without proper support.

Underutilisation: Without training, systems may fail to deliver expected ROI.

Migration Complexity: Moving 6.4GB of data and legacy integrations presents technical risks.

4. SaaS, PaaS & IaaS Cloud Computing Solutions

To modernise its infrastructure, CVS should adopt a hybrid cloud approach using SaaS, PaaS, and IaaS. This supports phased migration, closes operational gaps, and aligns with its budget and digital goals.

4.1 Software as a Service (SaaS)

SaaS provides ready-to-use cloud applications, ideal for CVS's distributed structure and limited IT resources. It reduces setup time, lowers maintenance, and enhances collaboration.

Recommended SaaS tools for CVS include:

- **Microsoft 365:** For cloud-based email, document collaboration, file sharing, and communication (Microsoft, 2022). Justification: Seamless integration with Azure ecosystem and nonprofit pricing makes it financially and technically viable.
- **Microsoft Dynamics 365 CRM & HR:** To manage donor relationships, volunteer records, payroll processing, and fundraising campaigns in one unified platform. Justification: Consolidates fragmented legacy systems, enhances donor retention, and streamlines HR processes.
- **Xero or QuickBooks Online:** To replace fragmented regional payroll software with a standard, cloud-based accounting solution. Justification: These tools provide compliance-ready financial reporting and automated payroll processing at low cost.

- **Mailchimp (with CRM integration):** For managing donor communications and automating email-based fundraising campaigns. Justification: Enables targeted communication and improves donor engagement through behavioural triggers and segmentation.
- **Square POS or Vend:** To upgrade point-of-sale operations across CVS's 260 charity shops, enabling real-time inventory tracking, digital receipts, and integrated donation checkout.

These solutions replace outdated tools, reduce overheads, and improve productivity. They are scalable and suited to CVS's remote-first model.

4.2 Platform as a Service (PaaS)

PaaS allows CVS to build and automate services without managing hardware, supporting both modern app development and legacy migration.

Recommended PaaS solutions:

- **Microsoft Azure App Services:** For custom-built internal dashboards, volunteer portals, and shop reporting tools. Justification: Allows rapid deployment and version control of secure web apps without infrastructure overhead.
- **Azure Logic Apps / Power Automate:** For low-code automation of routine workflows like onboarding or donation receipts. Justification: Reduces manual workload, improves response time, and enhances service consistency.
- **Azure Functions:** To create serverless functions for batch data processing, scheduled reports, and alert systems. Justification: Cost-effective, event-driven processing that scales automatically with demand.
- **Google Firebase:** As a lightweight, mobile-ready backend for any app-based donation or volunteer tracking system. Justification: Ideal for future mobile applications aimed at youth engagement and remote volunteering.

These PaaS offerings allow CVS to incrementally modernise specific services while reducing reliance on failing legacy databases. They also allow for agile experimentation without locking into heavy infrastructure.

4.3 Infrastructure as a Service (IaaS)

IaaS offers secure, flexible infrastructure for hosting legacy applications and managing data securely during the transition.

Recommended IaaS solutions:

- **Azure Virtual Machines:** For hosting any remaining custom-built or third-party applications that cannot yet be migrated to SaaS. Justification: Enables a “lift-and-shift” strategy with minimal reconfiguration, ensuring operational continuity.
- **Azure Blob Storage / Amazon S3:** To securely store donor, financial, and membership records with built-in redundancy and access control. Justification: Ensures compliance with GDPR and ISO 27001 while lowering data storage costs.
- **Azure Virtual Network and Firewalls:** For managing internal traffic and creating secure, segmented cloud environments. Justification: Supports network isolation, enhanced security, and zero-trust configurations.
- **Azure Backup & Site Recovery:** To ensure data resilience and disaster recovery. Justification: Reduces risk of data loss, supports RTO/RPO objectives, and enables regulatory compliance.

IaaS can bridge the gap between CVS’s current operational needs and its future ambitions for full digital transformation. It allows legacy infrastructure to operate securely while enabling phased migration.

Alignment with CVS Goals

This hybrid strategy will:

- **Reduce software licensing costs** through nonprofit-tiered services.
- **Eliminate fragmented systems**, enabling a single source of truth for all operational data.
- **Improve performance and reliability** through automated scaling and reduced downtime.
- **Support new digital services**, such as e-commerce, volunteer portals, and fundraising dashboards.
- **Enable secure and scalable infrastructure** that complies with legal and ethical responsibilities, including GDPR.

This cloud service model mix is strategically chosen to reflect CVS’s budgetary constraints, technical environment, and long-term sustainability ambitions.

5. Recommended Commercial Provider and Cost Estimates

Following detailed comparison and evaluation of major cloud providers—Amazon Web Services (AWS), Google Cloud Platform (GCP), IBM Cloud, and Microsoft Azure—this section justifies the selection of the most suitable commercial partner for CVS and presents a realistic cost estimate for implementation over a 3-year period.

5.1 Recommended Provider: Microsoft Azure

Microsoft Azure is the most appropriate cloud provider for CVS due to the following reasons:

- **Nonprofit Advantage:** Microsoft offers substantial discounts and donation programs for charitable organisations through the Tech for Social Impact initiative (Microsoft, 2022).
- **Integration with Existing Ecosystem:** CVS is expected to transition from Microsoft-based legacy tools (e.g., Excel, Outlook), making Microsoft 365 and Dynamics 365 natural successors with minimal retraining required.
- **UK-Based Data Centres:** Azure operates multiple data regions within the UK, supporting GDPR-compliant data sovereignty (Microsoft Trust Center, 2023).
- **Security and Compliance:** Azure offers 90+ compliance certifications, including ISO 27001, GDPR, and NHS DSPT, ensuring CVS can meet regulatory obligations.
- **Sustainability Commitment:** Azure is carbon neutral and aims to be carbon negative by 2030 (Microsoft Environmental Sustainability Report, 2023).

5.2 Cost Estimates

A conservative estimate has been developed for a 3-year transformation project within the allocated **£10 million** budget. All figures are approximate and based on 2024–2025 commercial pricing for nonprofits.

Service/Item	Estimated Cost (3 Years)
Microsoft 365 (Licensing for 4000 staff)	£1,200,000
Azure Virtual Machines, Backup & Storage	£2,500,000
Dynamics 365 CRM & HRM (SaaS)	£1,500,000

Service/Item	Estimated Cost (3 Years)
E-Commerce Portal Development (PaaS)	£1,000,000
INaaS/BPaaS Design and Rollout	£2,000,000
Staff Training, Change Management	£800,000
Cloud Consultancy, Migration & Testing	£500,000
Contingency (5%)	£500,000
Total	£10,000,000

Financial Benefits

This budget facilitates:

- A full system transition over 3 years.
- Long-term IT cost reduction (current costs are £25.7M annually).
- Elimination of costly support for legacy systems.
- Centralisation and automation of business processes.

According to IDC (2023), organisations that migrated to Microsoft Azure realised an average 36% reduction in IT infrastructure costs and a 56% increase in IT staff efficiency.

6. INaaS & BPaaS Development Recommendations

To fully leverage cloud transformation, CVS must not only migrate applications but also modernise its data strategy and operational workflows. This section presents tailored recommendations for implementing **Information as a Service (INaaS)** and **Business Process as a Service (BPaaS)** models over the next three years.

6.1 Information as a Service (INaaS)

INaaS facilitates centralised, real-time access to organisational data — enabling improved decision-making, transparency, and collaboration across CVS's operations.

Recommended INaaS initiatives:

1. **Centralised Cloud Data Lake**

Store donor, volunteer, and shop transaction records in Azure Data Lake or Amazon S3 to consolidate fragmented data sources (Raj & Kumar, 2020).

2. **Real-Time Analytics Dashboards**

Develop Power BI dashboards for fundraising, volunteer engagement, and donor lifecycle analytics (Microsoft, 2023).

3. **GDPR-Compliant Data Governance**

Use **Azure Purview** to catalogue, classify, and monitor sensitive data with full traceability and policy enforcement (Microsoft Trust Center, 2023).

4. **Master Data Management (MDM)**

Implement unified identity and profile management for members and donors using Microsoft's **Dataverse** or AWS Glue (IBM Cloud Docs, 2022).

5. **Information Self-Service Portals**

Enable staff to access real-time reports and dashboards without IT intervention — empowering local decision-making.

6.2 Business Process as a Service (BPaaS)

BPaaS allows CVS to streamline operations such as volunteer onboarding, donations, and e-commerce fulfilment by automating complex workflows across departments.

Recommended BPaaS initiatives:

1. **Volunteer Onboarding & Management Workflow:** Automate sign-ups, ID verification, training modules, and scheduling using Power Automate or AWS Step Functions.

2. **Donor Engagement Automation:** Configure personalised thank-you emails, tax receipts, and anniversary messages via CRM-integrated BPaaS logic (e.g. Dynamics 365 Marketing).
3. **E-Commerce Logistics Management:** Build a unified system to manage packaging, tracking, and dispatch of high-value items donated in shops — integrated with courier APIs and a central pricing tool.
4. **Fundraising Campaign Planning Tool:** Deploy a BPaaS-based interface for campaign design, budget approval, progress tracking, and KPI reporting.
5. **Performance Appraisal & Training:** Integrate BPaaS processes within the HRM platform to manage annual reviews, training outcomes, and goal setting.

7. Sustainability Discussion

Cloud migration offers CVS an opportunity to reduce environmental impact and align IT operations with its ethical values. Microsoft Azure, the recommended provider, leads the industry in sustainable practices.

Azure's data centres operate on 100% renewable energy (by 2025) and aim to be carbon negative by 2030 (Microsoft, 2023). Compared to traditional infrastructure, cloud platforms reduce energy use through higher server utilisation and optimised cooling (Accenture, 2020).

Azure also provides tools like the Emissions Impact Dashboard, enabling CVS to track and report its digital carbon footprint. This transparency supports eco-conscious donor engagement and reinforces CVS's social responsibility.

In summary, cloud adoption enhances CVS's operational efficiency while advancing its commitment to sustainability and the UN Sustainable Development Goals.

8. Conclusion

Christian Volunteer Services (CVS) must modernise its outdated IT infrastructure to improve efficiency, reduce costs, and enhance service delivery. Adopting a cloud-based strategy using SaaS, PaaS, IaaS—supported by INaaS and BPaaS—will enable CVS to unify systems, strengthen data use, and scale its operations effectively.

Microsoft Azure is the recommended provider due to its security, nonprofit support, and sustainability credentials. A phased migration within a £10 million budget can reduce IT spending, replace legacy risks, and support long-term innovation. Despite some transitional challenges, the benefits clearly align with CVS's mission and future growth.

9. References

Accenture (2020) *Cloud computing and sustainability: The environmental benefits of moving to the cloud*. [Online] Available from: <https://www.accenture.com> [Accessed 18 July 2025].

Ali, M., Khan, S.U. and Vasilakos, A.V. (2022) 'Security in cloud computing: Opportunities and challenges', *Information Sciences*, 305, pp.357–383.

Amazon Web Services (2024) *Nonprofit cloud solutions*. [Online] Available from: <https://aws.amazon.com/nonprofits/> [Accessed 18 July 2025].

Armbrust, M. et al. (2010) 'A view of cloud computing', *Communications of the ACM*, 53(4), pp.50–58.

Berman, S.J., Kesterson-Townes, L., Marshall, A. and Srivathsa, R. (2012) 'How cloud computing enables process and business model innovation', *Strategy & Leadership*, 40(4), pp.27–35.

Gartner (2024) *Magic quadrant for cloud infrastructure and platform services*. [Online] Available from: <https://www.gartner.com> [Accessed 18 July 2025].

Google Cloud (2023) *Environmental report*. [Online] Available from: <https://cloud.google.com/sustainability> [Accessed 18 July 2025].

Hill, T. and Westbrook, R. (1997) 'SWOT analysis: It's time for a product recall', *Long Range Planning*, 30(1), pp.46–52.

IBM Cloud (2022) *Master data management on cloud*. [Online] Available from: <https://www.ibm.com/cloud/mdm> [Accessed 18 July 2025].

IDC (2023) *The business value of Microsoft Azure*. [Online] Available from: <https://info.microsoft.com> [Accessed 18 July 2025].

InfoTech Research Group (2022) *SaaS vs. IaaS vs. PaaS decision matrix*. [Online] Available from: <https://www.infotech.com> [Accessed 18 July 2025].

KPMG (2023) *Cloud cost optimisation in the nonprofit sector*. [Online] Available from: <https://home.kpmg> [Accessed 18 July 2025].

Microsoft (2022) *Cloud for nonprofits: Product guide*. [Online] Available from: <https://nonprofit.microsoft.com> [Accessed 18 July 2025].

Microsoft (2023) *Environmental sustainability report*. [Online] Available from: <https://www.microsoft.com/en-us/sustainability> [Accessed 18 July 2025].

Microsoft (2024) *Azure pricing calculator*. [Online] Available from: <https://azure.microsoft.com/en-us/pricing/calculator/> [Accessed 18 July 2025].

Microsoft Trust Center (2023) *Compliance offerings*. [Online] Available from: <https://learn.microsoft.com/en-us/azure/compliance/> [Accessed 18 July 2025].

Microsoft Trust Center (2023) *Data residency and compliance*. [Online] Available from: <https://www.microsoft.com/en-us/trust-center> [Accessed 18 July 2025].

Peterborough City Council (2018) *Moves IT to AWS cloud*. [Online] Available from: <http://www.cloudpro.co.uk> [Accessed 18 July 2025].

Raj, P. and Kumar, B. (2020) *Cloud computing: Principles and paradigms*. 2nd ed. Hoboken: Wiley.

Salesforce (2024) *Nonprofit success pack*. [Online] Available from: <https://www.salesforce.org> [Accessed 18 July 2025].

Sitaram, D. and Manjunath, A. (2011) *Moving to the cloud: Developing apps in the new world of cloud computing*. Oxford: Elsevier.

Wang, L., Ranjan, R., Chen, J. and Benatallah, B. (2014) 'Cloud computing: A perspective study', *New Generation Computing*, 32(2), pp.137–146.